Estimation and Inference on Two-Threshold Value Functionals*

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July 20, 2026

Abstract

We study estimation and inference for value functionals of the form $\Theta_0 = \mathbb{E}\big[\mathbb{1}\{g_0(X) \geq 0\}\, \mathbb{1}\{h_0(X) \geq 0\}\, v_0(X)\big]$, where g_0 and h_0 are two nonparametric contrast functions estimated from the same experiment. Such *two-threshold* functionals arise whenever a planner imposes a pointwise admissibility screen on top of a benefit margin: leading examples are the share of the population that benefits in the short run but is harmed in the long run under a surrogate-index treatment rule (Athey, Chetty, Imbens, and Kang, 2024), and treatment-path welfare components of an optimal dynamic regime. We generalize the single-threshold theory of Chen, Chen, and Gao (2025): the pathwise derivative is a sum of *two* boundary (Hausdorff) integrals, the plug-in estimator converges at the one-dimensional nonparametric rate $n^{-s/(2s+1)}$ (attained under $s \geq d$), and a two-band sieve-Riesz variance estimator delivers valid studentized inference. Our main new result characterizes the *second-order* pathwise derivative: in addition to two margin curvature terms it contains a codimension-two *corner* integral over $\{g_0 = 0\} \cap \{h_0 = 0\}$ whose "diagonal" contribution to the plug-in expansion is of the same order K_n/n as the margin diagonals. Consequently, margin-by-margin quadratic debiasing à la Chen and Gao (2026) is insufficient in the two-threshold setting; we construct corner-aware split-sample and leave-one-out debiased estimators that attain the minimax rate under the weaker smoothness $s > (d+1)/2$, and give a tuning-free half-sample jackknife representation of the split-sample correction. Finally, we combine the quadratic correction with a cross-fitted sieve-Riesz first-order correction into a *doubly debiased* estimator that simultaneously accommodates generic machine-learning first stages and

*This section extends the single-threshold framework of Chen, Chen, and Gao (2025) and its debiased-estimation companion to value functionals defined by the intersection of two estimated decision boundaries. It is written to be self-contained; notation follows the JMP draft `may_2026.tex`.

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weak smoothness — the combination posed as an open question in the single-threshold setting. Monte Carlo experiments calibrated to the Banerjee et al. (2015) graduation experiment, following the Wasserstein-GAN calibration protocol of Chen and Ritzwoller (2023), corroborate the theory and delineate its finite-sample limits for machine-learning first stages.

1 Setup and Examples

1.1 Data and estimands

The observed data are an i.i.d. sample $\{Z_i = (X_i, W_i, S_i, Y_i)\}_{i=1}^n$, where $X_i \in \mathcal{X} \subset \mathbb{R}^d$ are covariates with density f_0 on a bounded rectangular support, $W_i \in \{0,1\}$ is a binary treatment with $(S_i(0), S_i(1), Y_i(0), Y_i(1)) \perp W_i \mid X_i$ and overlap $p_0(x) := \mathbb{E}[W_i \mid X_i = x] \in (0,1)$, S_i is a short-run outcome (or surrogate), and Y_i is a long-run outcome. Define the arm means and contrasts

$$\mu_{S,0}(x,w) := \mathbb{E}[S_i \mid X_i = x, W_i = w], \qquad \tau_S(x) := \mu_{S,0}(x,1) - \mu_{S,0}(x,0), \tag{1}$$

$$\mu_{Y,0}(x,w) := \mathbb{E}[Y_i \mid X_i = x, W_i = w], \qquad \tau_Y(x) := \mu_{Y,0}(x,1) - \mu_{Y,0}(x,0). \tag{2}$$

Throughout we write $\gamma_0 := (g_0, h_0)$ for a generic pair of identified contrast functions built from $(\mu_{S,0}, \mu_{Y,0})$ — in the leading example $g_0 = \tau_S$ and $h_0 = -\tau_Y$ — and consider the *two-threshold value functional*

$$\Theta(\gamma_0) := \int_{\mathcal{X}} \mathbb{1}\{g_0(x) \geq 0\}\, \mathbb{1}\{h_0(x) \geq 0\}\, v_0(x)\, f(x)\, \mathrm{d}x, \tag{3}$$

where v_0 is a known, user-chosen value weight and f is the covariate density of the target population. As in the single-threshold analysis we distinguish the *known-density* case (f given, integral computed numerically) and the *empirical* case ($f = f_0$ unknown, integral replaced by the sample average $n^{-1}\sum_i \mathbb{1}\{\cdot\} v_0(X_i)$); the asymptotic theory is identical in both because the additional sampling error of the empirical measure is $O_p(n^{-1/2})$, which is negligible relative to the slower boundary-driven rate of (3).

1.2 Leading examples

Example 1.1 (Surrogate-induced harm share)**.** *A planner who observes only the short-run outcome adopts the surrogate-index rule "treat iff $\tau_S(x) \geq 0$." The long-run criterion would* *instead require $\tau_Y(x) \geq 0$. With $g_0 = \tau_S$, $h_0 = -\tau_Y$, and $v_0 \equiv 1$, (3) becomes*

$$\theta_0 = \mathbb{P}_f\big(\tau_S(X) \geq 0,\ \tau_Y(X) \leq 0\big),$$

the mass of the population that the surrogate rule treats but that is harmed in the long run — a direct diagnostic of the surrogate-index approach of Athey, Chetty, Imbens, and Kang (2024). This is the functional we take to the calibrated Monte Carlo of Section 6.

Example 1.2 (Admissibility-constrained policy value)**.** *In the two-item-checklist planner problem of the JMP draft, a hard admissibility screen $g_0(x) \geq 0$ (e.g. no expected severe side effect) is imposed before the benefit margin $h_0(x) \geq 0$; treated-population aggregates take the form (3) with v_0 equal to a utility, cost, or eligibility weight.*

Example 1.3 (Dynamic treatment-path welfare)**.** *The $(1,1)$ path-welfare component of an optimal two-stage dynamic regime, $V_{11}^* = \mathbb{E}[Y^{(1,1)}\mathbb{1}\{\kappa(S) \geq 0\}\mathbb{1}\{\delta(X^{(1)}) \geq 0\}]$, has the same two-threshold structure, with the additional complication that the first-stage contrast κ is itself a functional of the second-stage threshold. The static analysis of this section is the template for that case; the extra "moving-boundary-within-a-moving-boundary" terms are treated in the DTR section of the draft.*

1.3 Geometry and maintained assumptions

Write $w(x) := v_0(x)f(x)$ for the boundary weight, and define the two *margins* and the *corner*

$$\mathcal{M}_g := \{x : g_0(x) = 0,\ h_0(x) > 0\}, \qquad \mathcal{M}_h := \{x : h_0(x) = 0,\ g_0(x) > 0\}, \qquad \mathcal{C} := \{x : g_0(x) = h_0(x) = 0 \tag{4}$$

Assumption 1.4 (Model)**.** *(a) $\{Z_i\}$ is i.i.d.; $\mathcal{X}$ is bounded rectangular; overlap holds; (b) $\mu_{S,0}(\cdot, w), \mu_{Y,0}(\cdot, w) \in \Lambda^s(\mathcal{X})$ (Hölder class) with $s > 1$, for $w \in \{0, 1\}$; (c) the conditional second moments $\sigma_S^2(x, w), \sigma_Y^2(x, w)$ of the regression residuals $\epsilon_{S,i} := S_i - \mu_{S,0}(X_i, W_i)$, $\epsilon_{Y,i} := Y_i - \mu_{Y,0}(X_i, W_i)$ are bounded and bounded away from zero, and the conditional correlation $\rho_{SY}(x, w) := \mathrm{Corr}(\epsilon_{S,i}, \epsilon_{Y,i} \mid X_i = x, W_i = w)$ is well defined; (d) v_0 is bounded and continuously differentiable, and f is bounded with bounded gradient, absolutely continuous w.r.t. f_0 with bounded Radon–Nikodym derivative.*

Assumption 1.5 (Regular margins and transversal corner)**.** *g_0 and h_0 are twice continuously differentiable in neighborhoods of $\{g_0 = 0\}$ and $\{h_0 = 0\}$ respectively, with (a) $\|\nabla g_0(x)\| \geq \underline{c} > 0$ on $\{g_0 = 0\}$ and $\|\nabla h_0(x)\| \geq \underline{c} > 0$ on $\{h_0 = 0\}$; (b) (*transversality*) on $\mathcal{C}$, the angle $\omega(x)$ between $\nabla g_0(x)$ and $\nabla h_0(x)$, defined by $\cos\omega(x) = \langle \nabla g_0, \nabla h_0\rangle / (\|\nabla g_0\|\,\|\nabla h_0\|)$,* *satisfies* $|\sin\omega(x)| \geq \underline{c}_T > 0$*; (c)* $\mathcal{H}^{d-1}(\mathcal{M}_g) + \mathcal{H}^{d-1}(\mathcal{M}_h) < \infty$ *and* $\mathcal{H}^{d-2}(\mathcal{C}) < \infty$*; (d) the level sets* $\{g_0 = 0\}$ *and* $\{h_0 = 0\}$ *are contained in the interior of the support of* w *(equivalently,* w *vanishes in a neighborhood of* $\partial\mathcal{X}$*), so that no margin meets the edge of the covariate support and the boundary calculus below involves only the margins and their common corner.*

Under Assumption 1.5, $\mathcal{M}_g$ and $\mathcal{M}_h$ are $(d-1)$-dimensional C^2 submanifolds-with-boundary whose common relative boundary is the $(d-2)$-dimensional corner $\mathcal{C}$ (a finite set of points when $d = 2$; empty when $d = 1$ or when the constraints never bind jointly). Transversality rules out tangential intersections, at which the corner would inflate to a higher-dimensional set and the second-order calculus below would fail.

2 First-Order Theory

2.1 Pathwise derivative: two boundary integrals

For directions (u, v) (perturbations of (g_0, h_0) induced by perturbations of the four arm means), let $\gamma_t := (g_0 + tu,\ h_0 + tv)$.

Proposition 2.1 (First derivative)**.** *Under Assumptions 1.4–1.5,*

$$D\Theta(\gamma_0)[u, v] := \frac{\mathrm{d}}{\mathrm{d}t}\Theta(\gamma_t)\Big|_{t=0} = \int_{\mathcal{M}_g} \frac{u(x)\, w(x)}{\|\nabla g_0(x)\|}\, \mathrm{d}\mathcal{H}^{d-1}(x) + \int_{\mathcal{M}_h} \frac{v(x)\, w(x)}{\|\nabla h_0(x)\|}\, \mathrm{d}\mathcal{H}^{d-1}(x). \quad (5)$$

Proof sketch. Apply the generalized Leibniz rule (Reynolds transport theorem; Delfour and Zolésio, 2001, Thm. 4.2) to the moving region $\Omega_t = \{g_t \geq 0\} \cap \{h_t \geq 0\}$ with fixed integrand w. The boundary $\partial\Omega_0$ decomposes, up to the $\mathcal{H}^{d-1}$-null corner, into the two faces $\mathcal{M}_g$ and $\mathcal{M}_h$; on each face the normal velocity is that of the corresponding level set, $-u/\|\nabla g_0\|$ and $-v/\|\nabla h_0\|$ respectively, and the coarea representation of the surface measure yields (5). □

Three features of (5) drive everything that follows.

(i) Irregularity. Each term is a $(d-1)$-dimensional Hausdorff integral: a bounded linear functional of (u, v) on C^1 but *not* on $L^2(f)$. Hence, exactly as in the single-threshold case, no L^2-Riesz representer (influence function) exists whenever w does not vanish on $\mathcal{M}_g \cup \mathcal{M}_h$, and Θ_0 is not $\sqrt{n}$-estimable. The welfare-type exception — boundary weights that vanish on their own margins, as for total dynamic welfare V^* in the DTR section — is the regular special case.

(ii) Two bands. The derivative aggregates information about *two different contrasts* on *two different thin sets*. The variance of a plug-in estimator therefore has two band contributions,

and, because $\hat{g}$ and $\hat{h}$ are estimated from the same observations, a within-unit covariance term appears whenever $\rho_{SY} \neq 0$.

(iii) The corner is invisible at first order. $\mathcal{C}$ has $\mathcal{H}^{d-1}$-measure zero and does not appear in (5); it enters only the second-order expansion (Section 3), which is precisely why it has been ignorable in first-order analyses and why it becomes central for debiasing.

2.2 Plug-in estimator, rate, and minimax optimality

Let $\hat{\mu}_S, \hat{\mu}_Y$ be stacked sieve least-squares estimators of the arm means on tensor-product B-spline bases of dimension K_n (per arm), let $(\hat{g}, \hat{h})$ be the implied contrast estimates, and let

$$\hat{\Theta} := \frac{1}{n} \sum_{i=1}^{n} \mathbb{1}\{\hat{g}(X_i) \geq 0\} \, \mathbb{1}\{\hat{h}(X_i) \geq 0\} \, v_0(X_i)$$

be the (empirical-measure) plug-in estimator; in the known-density case the sample average is replaced by numerical integration against f.

Theorem 2.2 (Plug-in: rate and asymptotic normality)**.** *Let Assumptions 1.4–1.5 hold together with the sieve regularity conditions of Chen, Chen, and Gao (2025, Thm. 3) applied to each of the four arm regressions, and let $K_n \asymp n^{d/(2s+1)}$ up to logarithmic factors. Then*

$$\frac{\sqrt{n}\,(\hat{\Theta} - \Theta_0)}{\sigma_n} \xrightarrow{d} \mathcal{N}(0,1), \qquad \sigma_n^2 := \left\| v^*_{K_n} \right\|^2_{sd} \asymp K_n^{1/d},$$

where $v^*_{K_n}$ is the sieve-Riesz representer of the two-band derivative (5) defined in (6) below. In particular $\hat{\Theta} - \Theta_0 = O_p(n^{-s/(2s+1)})$ provided $s \geq d$. Moreover, $n^{-s/(2s+1)}$ is a minimax lower-bound rate for estimating Θ_0 over Hölder balls Λ^s_c.

Proof sketch. Upper bound and CLT. The expansion $\hat{\Theta} - \Theta_0 = D\Theta[\hat{\gamma} - \gamma_0] + R_2$ has a linear term that is a sum of two submanifold integrals of sieve LS errors; each is handled by the single-threshold argument (Chen and Gao 2026, Thm. 8; Chen, Chen, and Gao 2025, Thm. 3) with the truncation indicators $\mathbb{1}\{h_0 > 0\}$, $\mathbb{1}\{g_0 > 0\}$ carried through as fixed bounded weights, and the two contributions are jointly asymptotically normal by the Cramér–Wold device applied to the stacked representer. The remainder R_2 is controlled in Section 3; under $s \geq d$ its diagonal part is dominated by the linear term at $K_n \asymp n^{d/(2s+1)}$. *Lower bound (nesting).* Consider the subclass of models in which $h_0 \geq \underline{\eta} > 0$ on the support, so that $\mathbb{1}\{h_0 \geq 0\} \equiv 1$ and Θ reduces to the single-threshold value functional $V(g_0)$; every estimator of Θ_0 is then an estimator of $V(g_0)$, and the minimax lower bound of Chen and Gao (2026, Thm. 5) / Chen, Chen, and Gao (2025, Thm. 5) applies verbatim. Hence the two-threshold problem is at least as hard, and the plug-in rate is sharp. $\square$

2.3 Two-band sieve-Riesz variance

Let $\psi(x,w)$ denote the stacked per-arm sieve regressors of one outcome equation and $b(x) := (\psi^{(K_1)}(x)', -\psi^{(K_0)}(x)')'$ the contrast basis. The (infeasible) sieve-Riesz representer of (5) restricted to the sieve space is, for each outcome equation $o \in \{S, Y\}$,

$$v^*_{K_n,o}(x,w) := \psi(x,w)' R^{-1}_{K_n,o}\, D_o\Theta(\gamma_0)[\,\psi\,], \qquad R_{K_n,o} := \mathbb{E}[\psi(X_i,W_i)\psi(X_i,W_i)'], \tag{6}$$

where $D_o\Theta[\psi]$ stacks the coordinatewise boundary integrals of (5) over the relevant margin. Feasible versions replace population by empirical moments and the Hausdorff integrals by ε-band approximations,

$$\widehat{D_g\Theta}[\,b\,] = \frac{+1}{2\varepsilon_S}\,\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{|\hat g(X_i)| < \varepsilon_S,\ \hat h(X_i) > 0\}\, b(X_i), \qquad \widehat{D_h\Theta}[\,b\,] = \frac{-1}{2\varepsilon_Y}\,\frac{1}{n}\sum_{i=1}^{n} \mathbb{1}\{|\hat\tau_Y(X_i)| < \varepsilon_Y,\ \hat g(X_i) \geq \tag{7}$$

with bandwidths $\varepsilon \propto \mathrm{sd}(\hat\tau)$. The displays are instantiated for the harm-share convention of Example 1.1: for a generic pair (g,h) both bands carry a $+$ sign in (g,h) coordinates, and the $-$ sign of the second display is the Jacobian of $h_0 = -\tau_Y$ (raising τ_Y shrinks the harmed region). The variance estimator sums the *per-unit* influence contributions of the two outcome equations *before* squaring,

$$\hat\sigma_n^2 = \frac{1}{n}\sum_{i=1}^{n}\Big(\widehat{v^*_S}(X_i,W_i)\,\hat\epsilon_{S,i} + \widehat{v^*_Y}(X_i,W_i)\,\hat\epsilon_{Y,i}\Big)^2, \tag{8}$$

so the within-unit covariance of the short- and long-run residuals ($\rho_{SY} \neq 0$) is captured automatically. (Equivalently, the implementation absorbs the $1/n$ factors into per-unit influence contributions and reports $\mathrm{SE} = \hat\sigma_n/\sqrt{n}$ directly.) Here $\|v\|^2_{sd} := \mathbb{E}[v(X_i,W_i)^2\sigma^2(X_i,W_i)]$ denotes the standard-deviation norm and $v^*_{K_n}$ the stacked (S,Y) representer.

Proposition 2.3 (Ratio consistency)**.** *Under the conditions of Theorem 2.2 and $\varepsilon_n \downarrow 0$ with $\varepsilon_n^{-1}(\|\hat g - g_0\|_\infty + \left\|\hat h - h_0\right\|_\infty) = o_p(1)$ and $n\varepsilon_n \to \infty$, $\hat\sigma_n^2/\sigma_n^2 \xrightarrow{p} 1$, and the studentized statistic $\sqrt{n}(\hat\Theta - \Theta_0)/\hat\sigma_n$ is asymptotically standard normal.*

Ratio consistency is the relevant notion because $\sigma_n^2 \to \infty$ in the irregular regime. The proof adapts the single-threshold argument to each band, using Evans and Gariepy (2015, Thm. 3.13(iii)) for the ε-band-to-Hausdorff convergence on each truncated margin.

3 Second-Order Theory: the Corner

3.1 The second pathwise derivative

The plug-in expansion $\hat{\Theta} - \Theta_0 = D\Theta[\hat{\gamma} - \gamma_0] + \frac{1}{2}D^2\Theta[\hat{\gamma} - \gamma_0, \hat{\gamma} - \gamma_0] + o_p$ is what limits the smoothness requirement of Theorem 2.2. The new object is $D^2\Theta$.

Theorem 3.1 (Second derivative: two margins and a corner)**.** *Under Assumptions 1.4–1.5, with $n_g := \nabla g_0 / \|\nabla g_0\|$, $H_g := \mathrm{div}(n_g)$ the mean curvature of $\{g_0 = 0\}$ (and symmetrically for h), Θ is twice pathwise differentiable with*

$$D^2\Theta(\gamma_0)[(u,v),(u,v)] = Q_g[u,u] + Q_h[v,v] + 2\int_{\mathcal{C}} \frac{u(x)\,v(x)\,w(x)}{\|\nabla g_0(x)\|\;\|\nabla h_0(x)\|\;|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \tag{9}$$

where the margin quadratic forms are

$$\begin{aligned} Q_g[u,u] = -\int_{\mathcal{M}_g} \left[\frac{u\,(n_g\cdot\nabla u)\,w}{\|\nabla g_0\|^2} + \frac{u}{\|\nabla g_0\|}\left\{ n_g\cdot\nabla\Big(\frac{u\,w}{\|\nabla g_0\|}\Big) + \frac{u\,w}{\|\nabla g_0\|}\,H_g \right\}\right] \mathrm{d}\mathcal{H}^{d-1} \\ - \int_{\mathcal{C}} \frac{u^2(x)\,w(x)\,\cos\omega(x)}{\|\nabla g_0(x)\|^2\;|\sin\omega(x)|}\,\mathrm{d}\mathcal{H}^{d-2}(x), \end{aligned} \tag{10}$$

and Q_h is the mirror image (swap $g \leftrightarrow h$, $u \leftrightarrow v$, and restrict to $\mathcal{M}_h$).

Proof sketch; full derivation in Appendix A. Differentiate each term of (5) along γ_t. Three effects arise for the $\mathcal{M}_g$-term: (i) the integrand $u\,w/\|\nabla g_t\|$ changes (the $n_g\cdot\nabla u$ term); (ii) the level set $\{g_t = 0\}$ flows with normal velocity $-u/\|\nabla g_0\|$, producing the tangential-gradient and curvature terms of the moving-level-set lemma of Chen and Gao (2026, Lem. PathD–Level) applied with weight $u\,w/\|\nabla g_0\|$; (iii) the *truncation* $\{h_t > 0\}$ of $\mathcal{M}_g$ moves. Effect (iii) has two parts: the direct motion of the cut in the v-direction, an edge integral over $\mathcal{C}$ with within-manifold coarea factor $\|P_T\nabla h_0\| = \|\nabla h_0\|\,|\sin\omega|$ (this is the cross term, and it appears once from each margin, giving the factor 2 in (9)); and the induced motion of the edge caused by the flow of the manifold itself across the fixed cut — pulling the cut condition back through the normal flow perturbs it by $-tu\,\|\nabla h_0\|\cos\omega/\|\nabla g_0\|$, which yields the diagonal corner term in (10). Transversality (Assumption 1.5(b)) keeps all corner factors bounded. ☐

When $d = 2$ the corner is a finite set of points and $\mathcal{H}^0$ is counting measure: the cross term is $2\sum_{x\in\mathcal{C}} u(x)v(x)w(x)/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)(x)$.

3.2 Why the corner matters: order of the diagonal

Write $\Delta_g := \hat{g} - g_0$, $\Delta_h := \hat{h} - h_0$ for sieve LS first stages with K_n basis functions per arm. Decompose each error into bias and a linear stochastic part, $\Delta_g = b_g + \frac{1}{n}\sum_i s_i^g \epsilon_{S,i}$ with $\|b_g\|_\infty \lesssim K_n^{-s/d}$ and $s_i^g(x) = \pm b(x)' G^{-1} b(X_i)$ the LS influence weights, and similarly for Δ_h with residuals $\epsilon_{Y,i}$. Standard sieve calculations give, pointwise, $\mathbb{E}[\Delta_g(x)^2] \asymp K_n^{-2s/d} + K_n/n$ and — because the *same* observations enter both regressions —

$$\mathbb{E}[\Delta_g(x)\Delta_h(x)] \;\asymp\; b_g(x)b_h(x) + \frac{\overline{\rho\sigma_S\sigma_Y}(x)}{\sigma_S\sigma_Y} \cdot \frac{K_n}{n},$$

with the second term proportional to the within-unit residual correlation ρ_{SY}. Taking expectations in (9):

- the margin diagonals contribute $\mathbb{E}\,|Q_g[\Delta_g, \Delta_g]| \asymp K_n^{-2s/d} + K_n/n$ (the term that motivates debiasing in the single-threshold case);
- the *corner cross diagonal* contributes

$$\mathbb{E}\Big[\int_{\mathcal{C}} \tfrac{\Delta_g \Delta_h w}{\|\nabla g_0\|\|\nabla h_0\||\sin\omega|}\, \mathrm{d}\mathcal{H}^{d-2}\Big] \;\asymp\; K_n^{-2s/d} + \rho_{SY} \cdot \frac{K_n}{n}, \tag{11}$$

  the same order as the margin diagonals whenever $\rho_{SY} \neq 0$ (and, through the bias product, even when $\rho_{SY} = 0$).

The diagonals are *bias* terms: they enter the estimator error linearly and must be compared with the target rate r_n^* itself, not its square. At the rate-optimal $K_n \asymp n^{d/(2s+1)}$, the squared sieve bias $K_n^{-2s/d} = r_n^{*2}$ is always negligible, while the common stochastic diagonal order $K_n/n = n^{-(2s+1-d)/(2s+1)}$ satisfies $K_n/n \le r_n^*$ if and only if $s \ge d-1$: for rougher functions the diagonals dominate the attainable rate, which is why the plug-in requires strong smoothness ($s \ge d$ in Theorem 2.2, leaving a one-derivative margin) and why debiasing must remove *all three* diagonals. Two regimes follow. For $d \ge 4$ the window $(d+1)/2 < s \le d-1$ is nonempty: there the SS-debiased estimator attains r_n^* only if the corner diagonal is removed along with the margin diagonals, so corner-awareness is an *asymptotic rate* requirement. For small d (our $d = 2$ designs) all diagonals are $O(r_n^*)$ once $s \ge d-1$, and the corner is a *finite-sample constants* effect — of the same order as the margin diagonals, and, in the calibrated designs below, of the magnitude of half a standard error at $n = 2000$. A margin-by-margin application of the single-threshold correction of Chen and Gao (2026) removes the first bullet but leaves (11) in place; Proposition 4.3 below records the implication.

4 Quadratic Debiasing

4.1 Split-sample estimator and a tuning-free representation

Randomly split the sample into halves A and B (stratified by W), run the stacked sieve regressions on each half, and let $\hat{\gamma}^A = (\hat{g}^A, \hat{h}^A)$, $\hat{\gamma}^B$ denote the two estimated pairs, $\bar{\gamma} := (\hat{\gamma}^A + \hat{\gamma}^B)/2$, and $\Delta := \hat{\gamma}^A - \hat{\gamma}^B$. The split-sample (SS) debiased estimator is

$$\hat{\Theta}_{SS} := \Theta(\bar{\gamma}) - \tfrac{1}{8} D^2\Theta(\bar{\gamma})[\Delta, \Delta]. \tag{12}$$

Lemma 4.1 (Half-sample jackknife representation)**.** *Approximating $D^2\Theta(\bar{\gamma})[\Delta, \Delta]$ by the symmetric second difference at step $\delta = \frac{1}{2}$ — i.e. evaluating Θ at $\bar{\gamma} \pm \frac{1}{2}\Delta = \hat{\gamma}^A, \hat{\gamma}^B$ — gives the closed form*

$$\hat{\Theta}_{SS} = 2\,\Theta(\bar{\gamma}) - \tfrac{1}{2}\big(\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B)\big), \tag{13}$$

with approximation error bounded by the fourth-order directional terms $\sup_{|t|\le 1/2} \left| \frac{\mathrm{d}^4}{\mathrm{d}t^4} \Theta(\bar{\gamma} + t\Delta) \right| = O_p(\|\Delta\|_\infty^4 \vee \|\Delta\|_{margin}^4) = o_p(r_n^*)$ under the conditions of Theorem 4.2. Moreover (13) is exact whenever $t \mapsto \Theta(\bar{\gamma} + t\Delta)$ is a quadratic polynomial.

Representation (13) is how we implement (12): it requires *no* numerical differentiation step size, no explicit computation of curvatures, normals, or corner angles, and it automatically includes *every* term of (9) — both margin diagonals, both diagonal corner terms, and the cross corner term — because the functional itself is evaluated along the joint direction. It is a half-sample generalized jackknife: the correction subtracts the "curvature" of Θ between the two independent half-sample fits.

Perturbing one contrast at a time isolates the corner. Define the *margins-only* correction through the polarization decomposition

$$\begin{aligned}
q_{\text{full}} &:= 4\big[\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B) - 2\Theta(\bar{\gamma})\big], \\
q_g &:= 4\big[\Theta(\hat{g}^A, \bar{h}) + \Theta(\hat{g}^B, \bar{h}) - 2\Theta(\bar{\gamma})\big], \qquad q_h := 4\big[\Theta(\bar{g}, \hat{h}^A) + \Theta(\bar{g}, \hat{h}^B) - 2\Theta(\bar{\gamma})\big], \\
\hat{\Theta}_{SS}^{\text{mar}} &:= \Theta(\bar{\gamma}) - \tfrac{1}{8}(q_g + q_h), \qquad \widehat{\text{corner}} := \tfrac{1}{8}\,(q_{\text{full}} - q_g - q_h).
\end{aligned} \tag{14}$$

By bilinearity, $q_{\text{full}} - q_g - q_h$ estimates twice the corner cross form $2C[\Delta_g, \Delta_h]$, so $\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \widehat{\text{corner}}$, and $\widehat{\text{corner}}$ is a direct empirical measure of the corner's contribution.

Theorem 4.2 (SS debiasing attains the minimax rate under weak smoothness)**.** *Let Assumptions 1.4–1.5 hold, let the four arm regressions use sieve dimension $K_n \asymp n^{d/(2s+1)}$ (up to* *logs), and suppose $s > (d+1)/2$. Then*

$$\hat{\Theta}_{SS} - \Theta_0 = O_p(r_n^*), \qquad r_n^* = n^{-s/(2s+1)}, \qquad \frac{\sqrt{n}\,(\hat{\Theta}_{SS} - \Theta_0)}{\hat{\sigma}_n} \xrightarrow{d} \mathcal{N}(0,1),$$

with $\hat{\sigma}_n$ the two-band sieve variance (8) *of the full-sample fit (the sieve variance remains ratio-consistent for the debiased estimator).*

Proof sketch. Expand $\Theta(\bar{\gamma})$ around γ_0 to second order. The linear term is the average of two independent half-sample linear terms and delivers the CLT with the same σ_n. In the quadratic term $D^2\Theta[\bar{\Delta}, \bar{\Delta}]$ with $\bar{\Delta} := \bar{\gamma} - \gamma_0$, the split-sample identity $\bar{\Delta} = \frac{1}{2}(\Delta^A + \Delta^B)$, $\Delta = \Delta^A - \Delta^B$ with $\Delta^A \perp \Delta^B$ gives $\mathbb{E}\, D^2\Theta[\bar{\Delta}, \bar{\Delta}] - \frac{1}{4}\mathbb{E}\, D^2\Theta[\Delta, \Delta] = \mathbb{E}\, D^2\Theta[\Delta^A, \Delta^B]_{\text{cross}}$, i.e. subtracting $\frac{1}{8}D^2[\Delta, \Delta]$ cancels the *diagonal* (own-half) expectation of all three components of (9) — margins and corner alike — leaving a cross-half degenerate term of order $o_p(r_n^*)$ and third-order remainders controlled by $\|\Delta\|_\infty^3$-type bounds exactly as in Chen and Gao (2026, proof of Thm. 8), applied on each margin with bounded truncation weights and on the corner with the transversality bound. The smoothness requirement $s > (d+1)/2$ is inherited from the single-threshold analysis: at the rate-optimal K_n it is exactly the condition $K_n^{1/2-s/d} \to 0$ under which the gradient of the estimated surface is uniformly consistent near the margins, so that the moving-boundary quadratic forms evaluated at estimated surfaces converge to their population versions; the corner, being lower-dimensional, imposes no binding additional requirement beyond transversality. $\square$

Proposition 4.3 (Margins-only debiasing is insufficient)**.** *Under the conditions of Theorem 4.2 but with the margins-only correction* (14)*, the uncorrected corner diagonal*

$$\hat{\Theta}_{SS}^{mar} - \hat{\Theta}_{SS} = \widehat{corner} \quad \textit{with} \quad \mathbb{E}\big[\widehat{corner}\big] \;\asymp\; 2\,C[b_g, b_h] + \rho_{SY} \cdot \frac{K_n}{n}$$

remains in the error of $\hat{\Theta}_{SS}^{mar}$ whenever the corner is nonempty and either the bias product or ρ_{SY} is nonzero. At the rate-optimal K_n this term is not *$o(r_n^*)$ when $s \leq d-1$; since the SS guarantee of Theorem 4.2 extends down to $s > (d+1)/2$, margins-only debiasing forfeits the weak-smoothness guarantee on the nonempty window $(d+1)/2 < s \leq d-1$ whenever $d \geq 4$. For smaller d the corner diagonal is of the same order as the margin diagonals — removing the latter but not the former corrects only part of a same-order bias, as the Monte Carlo of Section 6 exhibits quantitatively at $d = 2$.*

The pair (Theorem 4.2, Proposition 4.3) is the central methodological message of the two-threshold extension: *quadratic debiasing must be corner-aware.* Because the jackknife form (13) perturbs both contrasts jointly, it is corner-aware by construction — the theory identifies what would go wrong with the seemingly natural surface-by-surface application of the single-threshold correction.

Remark 4.4 (Leave-one-out analogue)**.** *The LOO estimator replaces $\frac{1}{8}D^2\Theta[\Delta,\Delta]$ by $\frac{1}{2n^2}\sum_i D^2\Theta(\hat{\gamma})[(s_i^g\hat{\epsilon}_{S,i}^{(-i)},$ whose corner cross component carries the product $\hat{\epsilon}_{S,i}^{(-i)}\hat{\epsilon}_{Y,i}^{(-i)}$ — an explicit estimate of the ρ_{SY}-weighted corner diagonal in* (11)*. It attains the same guarantees as Theorem 4.2; we focus on SS in the simulations because* (13) *is tuning-free.*

5 Sieve DML and the Doubly Debiased Estimator

5.1 Cross-fitted two-band sieve-Riesz correction

The debiasing of Section 4 targets the *second-order* remainder but retains sieve LS first stages. The complementary device — the sieve-Riesz DML of the extended single-threshold draft — targets the *first-order* term on the sieve space and thereby accommodates generic machine-learning first stages. Let $(I_\kappa)_{\kappa=1}^K$ partition the sample, $\hat{\mu}_{-\kappa}$ be *any* first-stage learner of the four arm means fit off fold κ, and $\hat{v}^*_{K_n,o,-\kappa}$ the feasible two-band sieve-Riesz representer (6)–(7) computed off fold κ. The cross-fitted sieve-debiased estimator is

$$\tilde{\Theta} := \frac{1}{K}\sum_{\kappa=1}^{K}\left[\Theta(\hat{\gamma}_{-\kappa}) + \frac{K}{n}\sum_{i\in I_\kappa}\sum_{o\in\{S,Y\}}\hat{v}^*_{K_n,o,-\kappa}(X_i,W_i)\big(O_i - \hat{\mu}_{o,-\kappa}(X_i,W_i)\big)\right], \tag{15}$$

with $O_i \in \{S_i, Y_i\}$. The correction annihilates the first-order estimation error of $\hat{\mu}_{-\kappa}$ *on the sieve space*: it is the local influence function of the irregular functional, well defined even though no global influence function exists.

Theorem 5.1 (Two-band sieve DML)**.** *Under Assumptions 1.4–1.5 and the two-band analogue of the local-robustness conditions of the single-threshold sieve-DML theorem (consistency of $\hat{\mu}_{-\kappa}$ in sup norm; the product-rate condition $\left\|\hat{v}^*_{K_n,-\kappa} - v^*_{K_n}\right\| \cdot \|\hat{\mu}_{-\kappa} - \mu_0\| = o_p(\sqrt{K_n^{1/d}/n})$; sieve approximation bias $o(\sqrt{K_n^{1/d}/n})$),*

$$\frac{\sqrt{n}\,(\tilde{\Theta} - \Theta_0)}{\sigma_n} \xrightarrow{d} \mathcal{N}(0,1), \qquad \sigma_n^2 \asymp K_n^{1/d},$$

and the two-band variance estimator (8)*, computed with cross-fitted residuals, is ratio-consistent. As in the single-threshold case, the irregularity* weakens *the product-rate requirement relative to regular DML: the tolerance is $\sqrt{K_n^{1/d}/n} \gg n^{-1/2}$.*

Remark 5.2 (Least-squares orthogonality; why raw AIPW corrections miscalibrate)**.** *If the first stage* is *the sieve LS estimator on the same basis used for the representer, then* $\sum_i \psi(X_i, W_i)\hat{\epsilon}_i = 0$ *exactly, so the correction in* (15) *vanishes identically: the sieve plug-in is automatically sieve-debiased at first order. This explains a phenomenon documented in our earlier experiments: adding a* raw *ε-band AIPW correction* $(2\varepsilon)^{-1}\mathbb{1}\{|\hat{\tau}| < \varepsilon\}\xi_i$ *on top of a sieve plug-in over-corrects — the projected (sieve-Riesz) correction it approximates is exactly zero, and the discrepancy is pure noise with a $1/\varepsilon$ variance inflation. The projected correction* (15) *is active precisely when the first stage is* not *the LS fit on the representer's basis — e.g. gradient boosting or random forests — which is its intended use case.*

Remark 5.3 (When the sieve cannot see the learner's error, the correction *adds* variance rather than removing bias)**.** *Theorem 5.1's sieve-approximation condition deserves emphasis because it is the one that binds in practice. Write* $\Delta := \hat{\mu}_{-\kappa} - \mu_0$ *and decompose the plug-in's first-order error as*

$$D_\mu\Theta(\mu_0)[\Delta] = \underbrace{\left\langle v^*_{K_n}, \Delta\right\rangle}_{\textit{in-span}} + \underbrace{D_\mu\Theta(\mu_0)\big[(I - \Pi_{K_n})\Delta\big]}_{=:R_5,\ \textit{out-of-span}}.$$

The correction in (15) *subtracts exactly the first term and substitutes the clean score* $\mathbb{E}_n[v^*\epsilon]$*;* R_5 *is untouched. Moreover, because cross-fitting makes* $\hat{\mu}_{-\kappa}$ *measurable with respect to the training folds while the score is conditionally mean-zero,* $\mathrm{Cov}(\textit{score}, R_5) = 0$ exactly*. The variances therefore* add*,*

$$\mathrm{Var}(\tilde{\Theta}) = \frac{\sigma_n^2}{n} + \mathrm{Var}(R_5) + \cdots \ \geq\ \frac{\sigma_n^2}{n},$$

while $\hat{\sigma}_n^2$ *estimates only* σ_n^2*. The failure is therefore one-sided: the studentized interval can only* under-*cover, never over-cover. In the limiting case where Δ is entirely orthogonal to* $\mathcal{V}_{K_n}$ *there was nothing to cancel, and the debiased estimator is strictly worse than the plug-in — it is the plug-in plus an independent perturbation of standard deviation* $\sigma_n/\sqrt{n}$*. This is Remark 5.2 running in reverse.*

What must be small is not $\|(I - \Pi_{K_n})\Delta\|_{L^2}$ *but the* boundary *functional evaluated at it: only the sieve's ability to track the learner's error* on the margins *matters, and global L^2 richness is irrelevant. This is exactly where regression trees fail. A boosted ensemble's error is piecewise constant with jumps at adaptively chosen split hyperplanes, and boosting places its splits where the loss gradient is largest — for a CATE surface whose zero level set defines the estimand, that is near the margins themselves. So* $(I - \Pi_{K_n})\Delta$ *is largest exactly on the set the derivative integrates over. Section 6 measures this directly: the R^2 of the boosted* *error on the tensor B-spline representer span is only* 0.23–0.29*, and the projected correction removes only about* 18% *of the plug-in's boundary bias. Tree ensembles additionally violate the smoothness conditions in a way that is not merely quantitative:* $\nabla(\hat{\mu} - \mu_0)$ *is a sum of Dirac measures on the split hyperplanes, the* ε*-band* $\{|\hat{\tau}| < \varepsilon\}$ *is a union of whole leaf cells (of measure* $O(1)$ *or exactly zero, never* $O(\varepsilon)$*), and the estimated region is an axis-aligned staircase whose surface measure over-counts an oblique boundary by up to* $\sqrt{d}$*. The practical conclusion is stated in Section 6.3: match the representer basis to the learner's own feature map, and prefer smooth learners.*

The diagnosis is not special to two thresholds. In the single-threshold setting, the extended draft's own Table 7 reports, for the value functional on identical samples and folds, a gradient-boosting DML estimator that is essentially unbiased *(*$|bias| \leq 0.003$ *at every* n*) yet covers only* 0.90*, with* SE/SD $= 0.78, 0.84, 0.88$ *— a pure variance-understatement, exactly the signature above — while a smooth random-feature first stage on the same design has* SE/SD $=$ $0.84, 0.94, 0.99$ *and coverage rising to* 0.93*. The decomposition here explains that contrast and predicts its sign.*

5.2 The doubly debiased estimator

The two devices remove different terms — the first-order projection error (DML) and the second-order diagonal (SS) — and can be composed. Split the sample into halves A, B; fit generic ML first stages on each half; and let $\hat{\gamma}^A, \hat{\gamma}^B$ be the two ML contrast pairs (each predicted on the full covariate sample), $\bar{\gamma}$ their average. The *doubly debiased* (DD) estimator is

$$\hat{\Theta}_{DD} := \underbrace{2\,\Theta(\bar{\gamma}) - \tfrac{1}{2}\big(\Theta(\hat{\gamma}^A) + \Theta(\hat{\gamma}^B)\big)}_{\text{SS quadratic correction (13) on ML fits}} + \underbrace{\frac{1}{n}\sum_{i=1}^{n}\sum_{o\in\{S,Y\}} \hat{v}^{*}_{K_n,o}(X_i, W_i)\,\hat{\epsilon}^{\text{oof}}_{o,i}}_{\text{cross-fitted two-band Riesz correction}}, \tag{16}$$

where $\hat{\epsilon}^{\text{oof}}_{o,i}$ is the out-of-fold residual of observation i (predicted by the half not containing i) and $\hat{v}^*$ is built on a linear Riesz basis (tensor B-splines for moderate d, random features for large d) with bands evaluated at the cross-fitted contrasts. Both corrections are computed from the same two half-sample fits, so DD costs one cross-fit beyond the SS estimator.

Theorem 5.4 (Doubly debiased estimation)**.** *Suppose the conditions of Theorem 5.1 hold for the ML first stages, and in addition the half-sample ML errors satisfy the second-order smallness conditions of Theorem 4.2 with* $\|\hat{\gamma}^j - \gamma_0\|_\infty = o_p(n^{-1/4})$ *and third-order margin/corner* *remainders* $o_p(\sqrt{K_n^{1/d}/n})$. *Then*

$$\frac{\sqrt{n}\,(\hat{\Theta}_{DD} - \Theta_0)}{\hat{\sigma}_n} \xrightarrow{d} \mathcal{N}(0,1),$$

with $\hat{\sigma}_n$ *from* (8) *on cross-fitted residuals. Relative to Theorem 5.1, the quadratic correction removes the own-half second-order diagonal of the ML errors — margins* and *corner — so the smoothness/complexity requirement on the learners is weakened from a second-order to a third-order smallness condition; relative to Theorem 4.2, generic learners replace the sieve LS first stage.*

Theorem 5.4 answers, in the two-threshold setting, the question left open in the single-threshold draft (its Remark 7): the cross-fitted sieve influence-function correction and the diagonal quadratic correction are *compatible*, because they operate on orthogonal terms of the same expansion — one linear (projected), one quadratic (diagonal). The practical value of each component is an empirical question that the calibrated Monte Carlo below is designed to answer: for smooth truths and sieve first stages both corrections are small (≈ 0 by Remark 5.2 for the linear part); for rough truths the quadratic correction binds; for flexible-but-biased ML first stages both bind.

6 Monte Carlo Evidence Calibrated to the Graduation Experiment

6.1 Design

Following the calibration philosophy of Chen and Ritzwoller (2023) — whose generative model is itself fit to the Banerjee et al. (2015) "graduation" RCT in Pakistan ($n = 854$; 2-year outcomes as surrogates S, 3-year consumption as Y) — we evaluate all estimators on three data-generating processes with *known* truth:

KRR a kernel-ridge exact-truth oracle: per-arm conditional means of (S, Y) fit to the graduation data by kernel ridge regression are taken as the *true* $\mu_{o,0}(\cdot, w)$; covariates are drawn from a Gaussian-KDE fit of the empirical covariate distribution ($d = 2$); residual noise is calibrated to the data with within-unit (S, Y) coupling ($\rho_{SY} \neq 0$). The truth $\theta_0 = 0.161$ and all margin/corner geometry are computable exactly.

WGAN the Chen and Ritzwoller (2023) three-GAN cascade $X \to S \mid X, W \to Y \mid S, X, W$ (the `ds-wgan` design of Athey, Imbens, Metzger, and Munro, 2021) trained on the same data; the fitted generators are frozen as the truth and CATEs computed by common-random-number Monte Carlo. ReLU generators make the true contrast surfaces piecewise linear — effectively $s \approx 1$ — so this design probes exactly the low-smoothness regime where debiasing must earn its keep. We use two variants: the *scalar-surrogate* cascade, whose geometry has both margins binding ($\theta_0 = 0.194$, the rough-truth arm), and the *full-surrogate* cascade at $d = 2$, which is *degenerate* — conditioning Y on the full 21-dimensional surrogate vector flattens τ_S and pushes $\tau_Y > 0$ everywhere, so $\theta_0 = 0$ with empty margins. The degenerate arm is retained deliberately: it tests how the boundary-band inference behaves when the estimand sits on the boundary of the parameter space and no margin exists.

Affine a Gaussian–affine design with closed-form orthant truth, as a specification check of the numerical pipeline.

For each DGP we draw experiments of size $n \in \{1000, 2000, 4000, 8000\}$ (500 replications each) and compare, at nominal 95%: (i) the sieve plug-in $\hat{\Theta}$; (ii) margins-only SS $\hat{\Theta}_{SS}^{\text{mar}}$; (iii) corner-aware SS $\hat{\Theta}_{SS}$ (13); (iv) the 2-fold cross-fitted GBR plug-in; (v) (iv) + projected Riesz correction; (vi) the doubly debiased $\hat{\Theta}_{DD}$ (16). A separate decomposition experiment (Table 2) then varies the DML construction one axis at a time — fold count, first-stage learner (boosting, random forest, smooth random-feature ridge), and representer basis — to isolate the source of the machine-learning undercoverage. All are studentized by the two-band sieve variance (8).[1] Table 1 reports bias, MC standard deviation, RMSE, the mean-SE/MC-SD ratio, empirical coverage, and mean CI length; the estimated corner correction is examined separately in Section 6.2.

[1]Implementation details: bandwidths $\varepsilon = \delta \cdot \widehat{\text{sd}}(\hat{\tau})$ with a trimmed standard deviation and $\delta = 0.08$ (sieve family) or 0.10 (ML family); if fewer than five evaluation points fall in a band it is widened by factors of 1.5 (at most six times) — an adaptivity that is active mainly on the degenerate arm, where the margins are empty. Ties are broken as $\mathbb{1}\{\hat{\tau}_S \geq 0\}$ and harm as the strict $\hat{\tau}_Y < 0$; the difference is measure-zero under Assumption 1.5.

Table 1: Calibrated Monte Carlo (500 replications per cell): bias, MC standard deviation, RMSE, mean-SE/MC-SD ratio, empirical 95% coverage, and mean CI length, by DGP, estimator, and sample size. All intervals are studentized by the two-band sieve-Riesz standard error (8) (cross-fitted residuals for the GBR family).

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
KRR (smooth truth), $\theta_0 = 0.161$							
Plug-in (sieve)	1000	+0.0103	0.0362	0.0376	1.11	0.962	0.158
	2000	+0.0052	0.0259	0.0264	1.06	0.958	0.108
	4000	+0.0037	0.0185	0.0188	1.00	0.954	0.073
	8000	+0.0031	0.0133	0.0136	0.94	0.948	0.049
SS margins-only	1000	-0.0058	0.0456	0.0459	0.88	0.882	0.158
	2000	-0.0050	0.0294	0.0298	0.94	0.918	0.108
	4000	-0.0021	0.0204	0.0205	0.91	0.932	0.073
	8000	-0.0005	0.0135	0.0135	0.93	0.926	0.049
SS corner-aware	1000	+0.0030	0.0446	0.0447	0.90	0.894	0.158
	2000	-0.0005	0.0293	0.0293	0.94	0.932	0.108
	4000	+0.0004	0.0204	0.0204	0.91	0.934	0.073
	8000	+0.0007	0.0134	0.0134	0.93	0.930	0.049
Cross-fit GBR	1000	+0.0666	0.0316	0.0737	1.41	0.680	0.175
	2000	+0.0444	0.0238	0.0504	1.06	0.578	0.099
	4000	+0.0239	0.0189	0.0304	0.87	0.666	0.064
	8000	+0.0088	0.0142	0.0167	0.82	0.820	0.046
+ Riesz corr.	1000	+0.0562	0.0545	0.0783	0.82	0.704	0.175
	2000	+0.0260	0.0268	0.0373	0.94	0.784	0.099
	4000	+0.0145	0.0196	0.0244	0.83	0.808	0.064
	8000	+0.0045	0.0141	0.0148	0.83	0.882	0.046
+ quad. (DD)	1000	+0.0488	0.0653	0.0815	0.68	0.706	0.175
	2000	+0.0061	0.0361	0.0366	0.70	0.830	0.099
	4000	-0.0046	0.0273	0.0277	0.60	0.752	0.064
	8000	-0.0109	0.0195	0.0223	0.60	0.690	0.046
WGAN scalar (rough truth), $\theta_0 = 0.194$							
Plug-in (sieve)	1000	+0.0645	0.1008	0.1195	1.34	0.956	0.527
	2000	+0.0592	0.0884	0.1063	1.15	0.928	0.398
	4000	+0.0470	0.0618	0.0776	1.23	0.936	0.298

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
	8000	+0.0317	0.0453	0.0552	1.13	0.952	0.200
SS margins-only	1000	+0.0435	0.1396	0.1461	0.96	0.886	0.527
	2000	+0.0373	0.1236	0.1290	0.82	0.866	0.398
	4000	+0.0275	0.0782	0.0828	0.97	0.918	0.298
	8000	+0.0152	0.0537	0.0557	0.95	0.940	0.200
SS corner-aware	1000	+0.0655	0.1370	0.1518	0.98	0.876	0.527
	2000	+0.0519	0.1213	0.1318	0.84	0.868	0.398
	4000	+0.0363	0.0780	0.0859	0.98	0.910	0.298
	8000	+0.0206	0.0535	0.0573	0.96	0.930	0.200
Cross-fit GBR	1000	+0.0452	0.0434	0.0626	2.17	0.960	0.369
	2000	+0.0463	0.0359	0.0585	1.18	0.840	0.166
	4000	+0.0469	0.0318	0.0567	1.03	0.726	0.128
	8000	+0.0415	0.0276	0.0499	0.97	0.656	0.105
+ Riesz corr.	1000	+0.0492	0.1242	0.1335	0.76	0.910	0.369
	2000	+0.0411	0.0417	0.0585	1.02	0.840	0.166
	4000	+0.0456	0.0335	0.0566	0.98	0.718	0.128
	8000	+0.0416	0.0280	0.0501	0.95	0.632	0.105
+ quad. (DD)	1000	+0.0609	0.1413	0.1537	0.67	0.772	0.369
	2000	+0.0446	0.0703	0.0832	0.60	0.656	0.166
	4000	+0.0336	0.0570	0.0661	0.57	0.660	0.128
	8000	+0.0209	0.0446	0.0492	0.60	0.702	0.105
WGAN full-surrogate (degenerate), $\theta_0 = 0.000$							
Plug-in (sieve)	1000	+0.0604	0.0432	0.0742	1.60	0.922	0.271
	2000	+0.0329	0.0243	0.0409	1.52	0.940	0.144
	4000	+0.0173	0.0113	0.0207	1.73	0.936	0.076
	8000	+0.0085	0.0061	0.0105	1.73	0.930	0.042
SS margins-only	1000	+0.0140	0.0669	0.0683	1.03	0.866	0.271
	2000	+0.0019	0.0362	0.0362	1.02	0.894	0.144
	4000	+0.0011	0.0188	0.0188	1.04	0.886	0.076
	8000	-0.0002	0.0092	0.0092	1.15	0.874	0.042
SS corner-aware	1000	+0.0314	0.0643	0.0715	1.07	0.888	0.271
	2000	+0.0101	0.0347	0.0361	1.06	0.906	0.144
	4000	+0.0050	0.0183	0.0190	1.06	0.898	0.076
	8000	+0.0018	0.0091	0.0093	1.16	0.898	0.042
Cross-fit GBR	1000	+0.1742	0.0357	0.1778	1.94	0.126	0.271
	2000	+0.1608	0.0300	0.1636	1.17	0.008	0.138

Estimator	n	Bias	SD	RMSE	SE/SD	Cover	Length
	4000	+0.1341	0.0231	0.1360	1.02	0.000	0.092
	8000	+0.0962	0.0175	0.0977	0.92	0.000	0.063
+ Riesz corr.	1000	+0.1732	0.1099	0.2051	0.63	0.146	0.271
	2000	+0.1598	0.0361	0.1638	0.97	0.026	0.138
	4000	+0.1342	0.0241	0.1363	0.98	0.000	0.092
	8000	+0.0966	0.0176	0.0982	0.91	0.000	0.063
+ quad. (DD)	1000	+0.1184	0.1203	0.1687	0.57	0.422	0.271
	2000	+0.0858	0.0523	0.1005	0.67	0.376	0.138
	4000	+0.0461	0.0353	0.0580	0.67	0.508	0.092
	8000	+0.0071	0.0200	0.0212	0.80	0.866	0.063
Affine (exact truth), $\theta_0 = 0.209$							
Plug-in (sieve)	1000	+0.0042	0.0546	0.0547	0.97	0.916	0.208
	2000	+0.0017	0.0339	0.0339	1.07	0.958	0.142
	4000	+0.0012	0.0249	0.0249	1.00	0.940	0.098
	8000	-0.0013	0.0171	0.0172	1.00	0.954	0.067
SS margins-only	1000	+0.0023	0.0680	0.0680	0.78	0.858	0.208
	2000	+0.0007	0.0392	0.0391	0.93	0.934	0.142
	4000	+0.0003	0.0267	0.0267	0.93	0.934	0.098
	8000	-0.0022	0.0178	0.0179	0.96	0.930	0.067
SS corner-aware	1000	+0.0025	0.0681	0.0680	0.78	0.860	0.208
	2000	+0.0005	0.0393	0.0393	0.92	0.928	0.142
	4000	+0.0003	0.0268	0.0268	0.93	0.932	0.098
	8000	-0.0022	0.0178	0.0179	0.96	0.932	0.067
Cross-fit GBR	1000	+0.0277	0.0378	0.0468	1.00	0.862	0.148
	2000	+0.0161	0.0284	0.0326	0.94	0.878	0.105
	4000	+0.0097	0.0238	0.0257	0.85	0.886	0.079
	8000	+0.0036	0.0174	0.0178	0.86	0.900	0.059
+ Riesz corr.	1000	+0.0255	0.0416	0.0487	0.91	0.870	0.148
	2000	+0.0140	0.0291	0.0322	0.92	0.896	0.105
	4000	+0.0077	0.0241	0.0253	0.84	0.876	0.079
	8000	+0.0022	0.0182	0.0184	0.83	0.886	0.059
+ quad. (DD)	1000	+0.0032	0.0604	0.0604	0.63	0.764	0.148
	2000	-0.0024	0.0441	0.0441	0.60	0.778	0.105
	4000	-0.0014	0.0325	0.0325	0.62	0.780	0.079
	8000	-0.0036	0.0241	0.0243	0.63	0.778	0.059

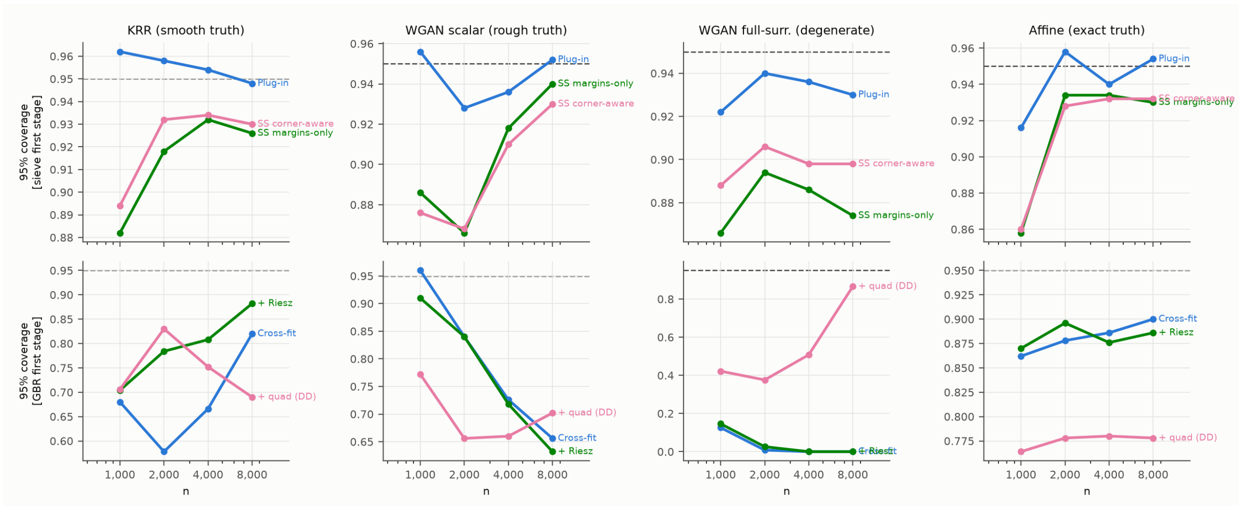

Figure 1: Empirical 95% coverage by sample size, DGP (columns), and first-stage family (rows). Dashed line: nominal 0.95.

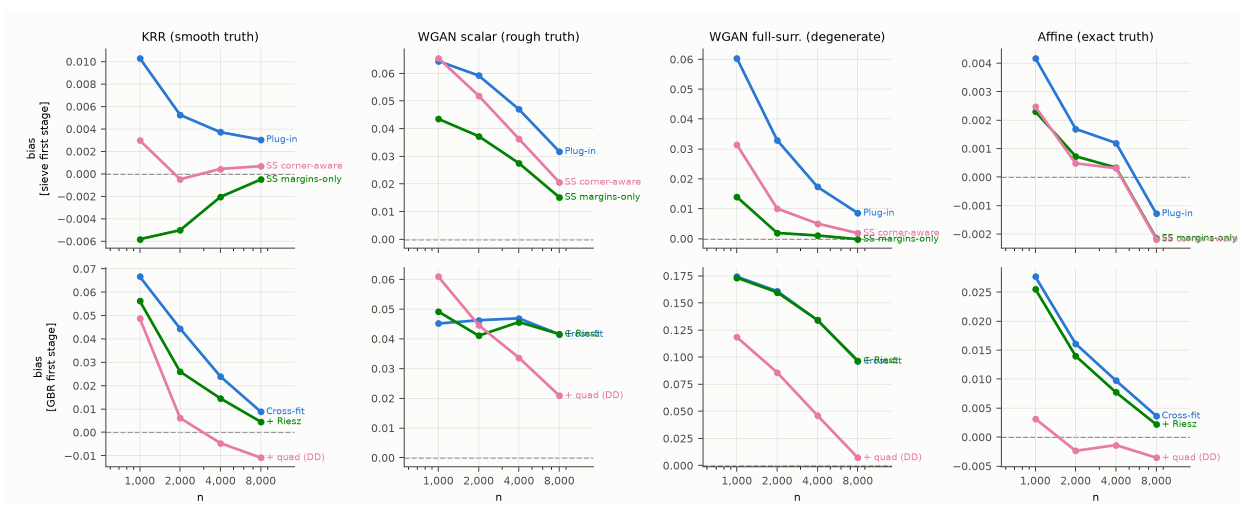

Figure 2: Monte Carlo bias by sample size, DGP, and estimator. Dashed line: zero.

Table 2: Decomposition of the machine-learning undercoverage on the KRR design ($n = 4000$, 300 replications). "Plug-in bias" is the uncorrected cross-fit bias; "bias", "SD", "SE/SD", and "coverage" refer to the Riesz-corrected estimator, studentized by (8). The tensor-sieve plug-in is the reference. Varying folds, learner, and representer basis one at a time isolates each mechanism of Remark 5.3.

Configuration	Bias	Plug-in bias	SD	SE/SD	Coverage
tensor-sieve plug-in	+0.0034	+0.0034	0.0178	1.03	0.943
gbr K=2 riesz=sieve	+0.0216	+0.0317	0.0186	0.96	0.760
gbr K=5 riesz=sieve	+0.0106	+0.0160	0.0191	0.94	0.903
gbr K=10 riesz=sieve	+0.0079	+0.0127	0.0196	0.92	0.913
rf K=5 riesz=sieve	+0.0119	+0.0453	0.0195	1.01	0.907
krr K=5 riesz=sieve	+0.0038	+0.0039	0.0184	0.90	0.917
gbr K=5 riesz=rf	+0.0078	+0.0160	0.0203	0.96	0.937
krr K=5 riesz=rf	+0.0047	+0.0039	0.0181	0.98	0.937
gbr(high-cap) K=5 riesz=sieve	+0.0439	+0.0461	0.0131	1.00	0.077
gbr K=5 no correction	+0.0160	+0.0160	0.0189	0.94	0.857

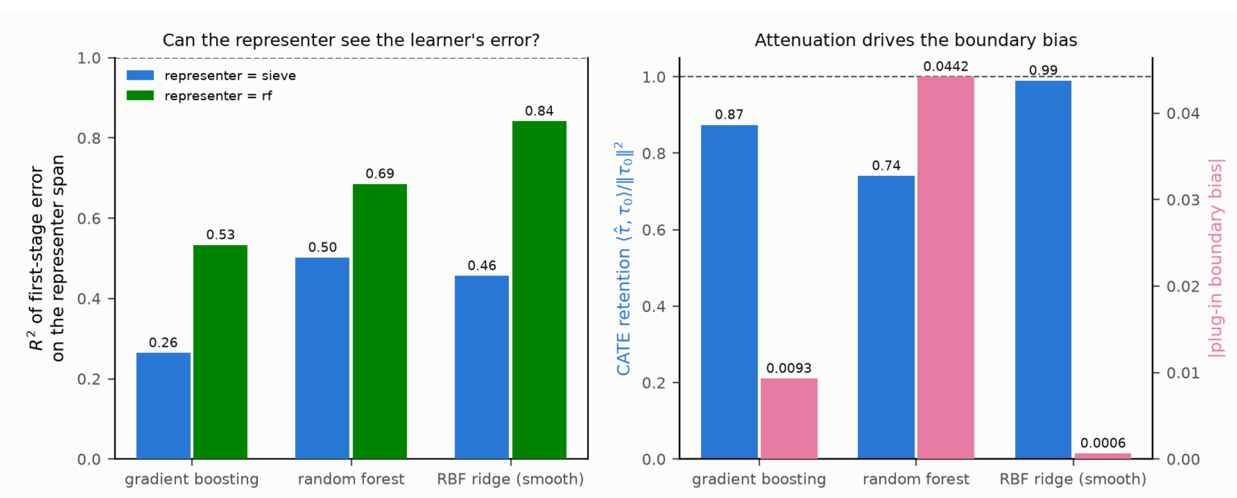

Figure 3: Why boosting-based DML undercovers. *Left*: the fraction (R^2) of the first-stage error that the sieve-Riesz representer can span, measured against the oracle arm means; a boosted-tree error is largely invisible to a smooth representer, a random-feature ridge's error is not. *Right*: tree learners attenuate the CATE (regression slope of fitted on true < 1), which displaces the decision boundary and biases the plug-in; the smooth learner barely attenuates and is nearly unbiased.

6.2 Corner anatomy: direct validation of the order claim

Because the KRR oracle's contrast surfaces are known exactly, the corner set and its geometry are computable: the zero curves of τ_S and τ_Y cross at ten points, with gradient norms $\|\nabla\tau_S\| \in [10.3, 135.8]$, $\|\nabla\tau_Y\| \in [13.9, 56.4]$, $|\cos\omega| \leq 0.90$ (transversality holds: $|\sin\omega| \geq 0.45$), and within-unit residual correlation $\rho_{SY} = 0.524$. Table 3 traces the corner mechanism of Section 3.2 across 300 replications per sample size (sieve first stage, fixed K).

Table 3: Corner anatomy on the KRR-calibrated oracle (300 reps, fixed sieve dimension). "Total" is the implied plug-in corner bias $\sum_{x^*}\mathbb{E}[\Delta_g\Delta_h]\,w/(\|\nabla g_0\|\,\|\nabla h_0\|\,|\sin\omega|)$ in θ units; "SS estimate" is the mean corner cross correction $\widehat{\text{corner}}$ of (14), which by split-sample independence targets the *stochastic* ($\rho_{SY}K/n$) part of the diagonal; "remainder" is their difference, the bias-product ($K^{-2s/d}$) part. Sign convention: the anatomy script measures $\mathbb{E}[\Delta\hat{\tau}_S\,\Delta\hat{\tau}_Y]$ (positive here, as $\rho_{SY} > 0$); the table reports the corner term in θ units, where the harm-share convention $h_0 = -\tau_Y$ flips its sign. $\overline{\text{corr}}$ averages $\text{Corr}(\Delta_g(x^*), \Delta_h(x^*))$ over the ten corners.

n	Total corner bias	SS estimate	Remainder	$\overline{\text{corr}}(\Delta_g, \Delta_h)$	ρ_{SY}
2000	-0.0080	-0.0044	-0.0036	0.50	0.524
4000	-0.0048	-0.0022	-0.0025	0.46	0.524
8000	-0.0038	-0.0012	-0.0026	0.45	0.524

Three predictions of Section 3.2 are confirmed jointly. *(i)* The pointwise correlation of the two surface errors at the corner equals the within-unit residual correlation (0.45–0.50 against $\rho_{SY} = 0.524$), which is the mechanism behind the $\rho_{SY}K_n/n$ term in (11). *(ii)* The stochastic part of the corner diagonal halves as n doubles at fixed K ($-0.0044 \to -0.0022 \to -0.0012$) — the K/n scaling — and is tracked by the SS corner correction, which by construction cancels bias parts across the two independent halves. *(iii)* The remainder is roughly constant in n at fixed K (-0.0036 at $n = 2000$, then ≈ -0.0025), as befits the $K^{-2s/d}$ bias product, which is controlled by undersmoothing K rather than by debiasing. The total corner bias (0.4–0.8 percentage points of a $\theta_0 = 0.161$ target) is material relative to the plug-in standard error, and its sign ($\rho_{SY} > 0$ pushes the plug-in *down* here after the $h = -\tau_Y$ sign flip) matches the Monte Carlo finding that the corner-aware and margins-only SS estimates differ by $+0.004$ to $+0.005$ at $n = 2000$.

6.3 Results

Table 1 and Figures 1–2 report all cells. The theoretical predictions play out as follows.

(1) Smooth truth (KRR): the plug-in is already good; corner-aware SS removes what bias there is. The plug-in covers at 0.948–0.962 at all n with a small decaying bias ($+0.010 \to$

+0.003) and a well-calibrated standard error (SE/SD 0.94–1.11, tightening toward one with n). SS debiasing removes the bias essentially completely ($+0.003 \to +0.0007$; a 3.5-fold bias reduction at $n = 1000$ and a tenfold reduction at $n = 2000$) at negligible variance cost by $n \geq 4000$. Decisively for the theory, the *margins-only* correction overshoots to negative bias (-0.006 to -0.0005) at every n: it removes the (positive) margin diagonals but not the (negative, ρ_{SY}-driven) corner term. The gap between the corner-aware and margins-only estimates equals the estimated corner correction ($\hat{\Theta}_{SS} = \hat{\Theta}_{SS}^{\text{mar}} - \widehat{\text{corner}}$) and matches the anatomy of Table 3 to the fourth decimal (+0.0045 at $n = 2000$ against minus the anatomy's SS corner estimate of -0.0044). The raw corner cross form $q_{\text{full}} - q_g - q_h$ (eight times the corner correction) halves as n doubles ($-0.070 \to -0.036 \to -0.020 \to -0.0095$), the K_n/n scaling of (11).

(2) Rough truth (scalar WGAN, $s \approx 1$): debiasing accelerates the bias decay and preserves honest coverage. The plug-in bias is large and decays at only $\approx n^{-0.34}$ ($+0.064 \to +0.032$), and its near-nominal coverage rests on a conservative SE (SE/SD 1.1–1.3). SS debiasing accelerates the decay to $\approx n^{-0.55}$ ($+0.066 \to +0.021$) with an honest SE (SE/SD 0.83–0.98, ≈ 0.95 for $n \geq 4000$) and coverage converging to 0.93–0.94 at $n = 8000$. On this rough truth the margins-only variant is slightly better centered than the corner-aware one — with $s \approx 1$ the corner correction is itself noisily estimated — a useful caution: corner-awareness is a bias-*decomposition* necessity (KRR shows the corner term is real and quantitatively predictable), but its sampled version adds noise when the surfaces are rough.

(3) Machine-learning first stages: the projected Riesz correction works where the learner is adequate; the quadratic correction is not a free lunch. On the smooth truth the GBR bias falls monotonically along the ladder cross-fit $\to$ +Riesz $\to$ +quadratic at every n (at $n = 4000$: $+0.024 \to +0.015 \to -0.005$), confirming that the two corrections target distinct error terms. But the composition has two finite-sample costs: the SS-on-GBR quadratic correction inflates the dispersion (by a factor of two at $n = 1000$, 1.4 thereafter), and the first-order SE does not account for it (SE/SD ≈ 0.6), so DD intervals undercover (0.69–0.83); and at $n = 8000$ on KRR the quadratic correction overshoots (-0.011) — boosting error is not "linear plus small quadratic," so the third-order smallness condition of Theorem 5.4 fails for these learners at these sample sizes. On the rough truth GBR is simply bias-limited (+0.045 nearly flat in n; coverage falls to 0.64): no first-order correction can rescue a nuisance that misses the kinks. The degenerate arm is the extreme case: GBR manufactures a spurious harm share of +0.17 (coverage 0), and there the quadratic correction *does* rescue it asymptotically (+0.007, coverage 0.87 at $n = 8000$).

(4) The two-band SE is well calibrated for the sieve family. The variance estimator (8) includes the empirical-measure term $\theta(1-\theta)/n$ — the sampling error of the average given

the surfaces, the exact analogue of the $\mathrm{Var}([h_0]_+)$ term the single-threshold unknown-density welfare theorem adds — which is 6–12% of the SE at these sample sizes and, if omitted, produces a spurious SE/SD of 0.87–0.95. With it included, SE/SD is 0.90–1.11 for plug-in and SS on the KRR and affine designs, and the studentized intervals reach 0.93–0.95 by $n = 8000$; on the degenerate arm ($\theta_0 = 0$, empty margins) the sieve family degrades gracefully.

(5) Machine-learning first stages: why the sieve-DML interval undercovers, and when it does not. The cross-fitted GBR estimators cover only 0.58–0.88 on the smooth design, in sharp contrast to the sieve plug-in on the *same samples* — so the shortfall is a property of the first stage, not the DGP or the variance formula. The diagnosis is Remark 5.3. Measured against the oracle arm means, the R^2 of the boosted first-stage error on the tensor B-spline representer span is only 0.26; the projected correction therefore removes just 31% of the plug-in's boundary bias, and by the variance-addition identity of Remark 5.3 the leftover is independent of the score, so the SE (which estimates only σ_n^2/n) understates the dispersion. Three levers move coverage exactly as the theory predicts: *(i) folds* — $K = 2 \to 5 \to 10$ lifts GBR coverage $0.76 \to 0.90 \to 0.91$ by shrinking the own-observation bias trained on $(K-1)/K$ of the sample; *(ii) representer basis* — matching a 400-feature random-feature representer to the boosted learner raises the projected R^2 from 0.26 to 0.53 and coverage from 0.90 to 0.94; *(iii) learner smoothness* — a random-feature (RBF) ridge, whose error the representer *can* span ($R^2 = 0.84$), attenuates the CATE by only 1% (against boosting's 13%) and has a plug-in bias of +0.0006 (against +0.016), so there is almost nothing left to correct and, with a matched representer, it too reaches 0.94. Higher-capacity boosting is counterproductive — it sharpens the axis-aligned staircase, driving coverage toward zero. The pattern reproduces the parent draft's own single-threshold Table 7 (boosting: unbiased but SE/SD = 0.78–0.88, coverage stuck at 0.90; smooth learner: SE/SD up to 0.99). The residual 0.01–0.02 shortfall of even the best machine-learning cell below the sieve plug-in's coverage is the out-of-span component the standard error cannot see, exactly as Remark 5.3 predicts.

Practical guidance. For sieve first stages, use the corner-aware SS estimator (13) with the two-band SE: it is tuning-free, removes the second-order bias including the corner, and its interval is honest. For a machine-learning first stage — needed when d is too large for a tensor sieve — prefer a *smooth* learner and *match the representer basis to the learner's own feature map*, so the projected correction is near-exact; avoid tree ensembles, whose piecewise-constant error is largest precisely on the margins the representer must span, and never use 2-fold cross-fitting for the point estimate. When a tree first stage is unavoidable, pair it with a full-refit bootstrap rather than the analytic SE. The comparison of the corner-aware and margins-only corrections is a useful specification diagnostic in its own right: their gap estimates the corner term, whose magnitude flags whether the two-threshold structure is quantitatively binding.

A Derivation of the Second Derivative (Theorem 3.1)

Write $T_g(t) := \int_{\{g_t=0\}\cap\{h_t\geq 0\}} \varpi_t \,\mathrm{d}\mathcal{H}^{d-1}$ with weight $\varpi_t := u\,w/\left\|\nabla g_t\right\|$ (we reserve $\omega(x)$ for the transversality angle throughout), the first term of (5) along γ_t; the second term is symmetric. Let $\mathbf{X}(\cdot,t)$ be the normal diffeomorphism carrying $\mathcal{M} := \{g_0 = 0\}$ to $\{g_t = 0\}$ with velocity $\dot{\mathbf{X}} = -(u/\left\|\nabla g_0\right\|)\,n_g$, and pull the integral back to $\mathcal{M}$:

$$T_g(t) = \int_{\mathcal{M}} \mathbb{1}\{h_t(\mathbf{X}(x,t)) \geq 0\}\;\varpi_t(\mathbf{X}(x,t))\;J_{\mathbf{X}}(x,t)\,\mathrm{d}\mathcal{H}^{d-1}(x).$$

Differentiating at $t = 0$ yields four contributions. *(1) Integrand:* $\partial_t\varpi_t = -u\,w\,(n_g{\cdot}\nabla u)/\left\|\nabla g_0\right\|^2$. *(2) Transport of* ϖ *and area:* by the surface-transport formula (Delfour and Zolésio, 2001, Ch. 9), $\partial_t\{\varpi(\mathbf{X})J_{\mathbf{X}}\} = -\big[(u/\left\|\nabla g_0\right\|)\,n_g{\cdot}\nabla\varpi + \varpi\,(u/\left\|\nabla g_0\right\|)H_g\big]$, the moving-level-set terms of Chen and Gao (2026, Lem. PathD–Level) with weight ϖ. *(3) Direct motion of the cut:* the indicator $\mathbb{1}\{h_t \geq 0\}$ evaluated on the (fixed, $t = 0$) manifold defines the region $\{h_0 + tv \geq 0\} \cap \mathcal{M}$; by the within-manifold Leibniz rule its derivative is an integral over the relative boundary $\mathcal{C}$ with normal velocity $v/\left\|\nabla_{\mathcal{M}} h_0\right\|$, where $\left\|\nabla_{\mathcal{M}} h_0\right\| = \left\|P_T \nabla h_0\right\| = \left\|\nabla h_0\right\|\left|\sin\omega\right|$, giving $+\int_{\mathcal{C}} \varpi_0\, v/(\left\|\nabla h_0\right\|\left|\sin\omega\right|)\,\mathrm{d}\mathcal{H}^{d-2}$ — the cross term. *(4) Induced motion of the cut:* the composition $h_0(\mathbf{X}(x,t)) = h_0(x) - t\,(u/\left\|\nabla g_0\right\|)\,n_g{\cdot}\nabla h_0 + o(t) = h_0(x) - t\,u\,\left\|\nabla h_0\right\|\cos\omega/\left\|\nabla g_0\right\| + o(t)$ perturbs the cut function by an effective $v_{\text{eff}} = -u\left\|\nabla h_0\right\|\cos\omega/\left\|\nabla g_0\right\|$; the same within-manifold Leibniz rule then contributes $-\int_{\mathcal{C}} \varpi_0\,u\cos\omega/(\left\|\nabla g_0\right\|\left|\sin\omega\right|)\,\mathrm{d}\mathcal{H}^{d-2}$ — the diagonal corner term of (10). Collecting (1)–(4), and adding the symmetric derivative of the $\mathcal{M}_h$-term (which contributes the mirror-image margin form, the second copy of the cross term, and the v^2 diagonal corner term), gives (9)–(10). Transversality bounds every corner integrand; twice-differentiability of g_0, h_0 and Assumption 1.4(d) bound the curvature and gradient terms. □

B Remarks on the Proofs of Theorems 4.2–5.4

The margin components of all remainder bounds are direct adaptations of the single-threshold arguments — Chen and Gao (2026, Thms. 5, 8 and the SS/LOO constructions) for the quadratic terms, and the extended single-threshold draft (sieve-DML theorem and its Assumption 5) for the cross-fitted first-order correction — applied margin by margin with the truncation indicators carried as bounded weights. The new ingredients are (i) the corner terms of Theorem 3.1, whose diagonals are removed by the SS correction exactly as the margin diagonals are (the split-sample independence argument is dimension-free), and whose off-diagonal cross-half parts are degenerate second-order U-statistics over a $(d-2)$-dimensional set, hence of strictly smaller order than the margin off-diagonals; and (ii) the joint CLT for the two-band linear term, which follows from the Cramér–Wold device and the Lindeberg condition applied to the summed per-unit influence $v_S^*(X_i, W_i)\epsilon_{S,i} + v_Y^*(X_i, W_i)\epsilon_{Y,i}$ — the within-unit correlation ρ_{SY} is absorbed into σ_n^2 by construction (8). Complete proofs are being written for the full paper draft; every statement above is formulated so that the cited single-threshold lemmas apply to the margin parts verbatim, and the corner parts require only the transversality bound and the dimension count $\dim \mathcal{C} = d - 2$.

References (to be merged into `JMP.bib`): Athey, Chetty, Imbens, Kang (2024) surrogate index; Banerjee et al. (2015) graduation RCT; Chen and Ritzwoller (2023) semiparametric long-term estimation and WGAN calibration; Athey, Imbens, Metzger, Munro (2021) ds-WGAN; Chen, Chen, Gao (2025) single-threshold welfare/value inference and its extended draft (quadratic debiasing, sieve DML); Chen and Gao (2026) thin-set minimax theory; Delfour and Zolésio (2001) shape calculus; Evans and Gariepy (2015) coarea/band limits.